MS650A - 2S24

Exercícios da aula de 30/10

(1) Resolva os seguintes problemas:

$$\text{(a)}\ \begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = 8 \operatorname{sen}^2 x, \\ u(0,t) = 0, \quad u(\pi,t) = 0. \end{cases}$$

$$\text{(b)}\ \begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < \pi, \quad t > 0, \\ u(x,0) = x^3, \quad u_t(x,0) = 0, \\ u(0,t) = 0, \quad u_x(\pi,t) = 0. \end{cases}$$

$$\text{(c)}\ \begin{cases} u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = 0, \\ u(0,t) = t, \quad u(l,t) = 1. \end{cases}$$

$$\text{(d)}\ \begin{cases} u_{tt} - u_{xx} = x - t, \quad 0 < x < 1,\ t > 0, \\ u(x,0) = x, \quad u_t(x,0) = x^2/2, \\ u(0,t) = t, \quad u(1,t) = t^2/2. \end{cases}$$

$$\text{(e)}\ \begin{cases} u_{tt} = c^2 u_{xx} + \Omega, \quad 0 < x < l,\ t > 0, \\ u(x,0) = 0, \quad u_t(x,0) = 0, \\ u(0,t) = 0, \quad u(l,t) = 0, \quad \Omega = \text{cte} \end{cases}$$

$$\text{(f)}\ \begin{cases} u_t = 100 u_{xx}, \quad 0 < x < 10,\ t > 0, \\ u(x,0) = 0, \\ u(0,t) = 10, \quad u(10,t) = 30. \end{cases}$$

$$\text{(g)}\ \begin{cases} u_t + 4(u-5) = u_{xx},\ 0 < x < 1,\ t > 0, \\ u(x,0) = 5 + x, \\ u(0,t) = 5, \quad u(1,t) = 5. \\ \text{sugestão: } u(x,t) = A + \mathrm{e}^{Bt} v(x,t) \end{cases}$$

(2) Resolva, usando coordenadas cartesianas, $u = u(x,y,z)$:

$$\begin{cases} \nabla^2 u = 0, \quad 0 < x < a,\ 0 < y < b,\ 0 < z < c, \\ u(0,y,z) = \operatorname{sen} \frac{\pi y}{b} \operatorname{sen} \frac{\pi z}{c}, \quad u(a,y,z) = 0, \\ u(x,0,z) = u(x,b,z) = u(x,y,0) = u(x,y,c) = 0. \end{cases}$$